
Research on the Construction of a Data-Driven Evaluation System for General Artificial Intelligence Courses in Higher Vocational Education

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Abstract. With the rapid advancement of artificial intelligence (AI) technology, the education sector is undergoing profound transformations. The teaching quality of general AI courses in higher vocational colleges is pivotal to the improvement of students' digital literacy and their long-term career development. Nevertheless, the current evaluation systems for such courses suffer from prominent drawbacks, including single evaluation modes, delayed feedback mechanisms, the absence of dynamic learning monitoring, and insufficient personalized evaluation design. Based on the theory of multiple intelligences, this study constructs a full-process evaluation system covering pre-class, in-class and post-class teaching stages. Preliminary evaluation indicators are obtained through expert questionnaires, and key indicators are extracted via dimensionality reduction using the principal component analysis (PCA) method. Furthermore, the grey wolf optimization (GWO) algorithm is adopted to optimize the hyperparameters of the random forest model for determining the weight of each evaluation indicator, which effectively improves the accuracy and efficiency of course evaluation. The research findings can provide a scientific basis and practical guidance for the quality improvement of general AI courses in higher vocational colleges, and support the personalized development and comprehensive ability cultivation of students.

Keywords: general artificial intelligence courses; evaluation system; multiple intelligences; principal component analysis; grey wolf optimization algorithm

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1. Introduction

The rapid development of AI technology is profoundly reshaping the educational ecosystem. At the national level, policy documents such as the Action Plan for Artificial Intelligence Innovation in Institutions of Higher Education have been issued to promote the in-depth integration of artificial intelligence and educational teaching and build a talent training system adapted to the intelligent era. Tongji University also released the Action Plan for Disciplinary Innovation and Development Empowered by Artificial Intelligence (2024-2027), putting forward clear requirements for educational innovation in the AI era. As the core platform for cultivating high-quality technical and skilled talents, higher vocational colleges undertake the key mission of improving students' digital literacy and adapting to industrial upgrading through general AI education. The scientificity of the teaching quality evaluation system for general AI courses directly determines the realization of talent training objectives.

However, the research on the evaluation of general AI courses in higher vocational colleges is still in the initial stage, with several prominent problems in current practical evaluation systems. First, existing evaluation methods focus excessively on summative assessment while ignoring the process-based monitoring of students' practical abilities. Second, the design of evaluation indicators is disconnected from professional development needs, lacking consideration of students' interdisciplinary application capabilities. Third, the lagging feedback mechanism fails to support targeted personalized teaching improvement. These deficiencies are inconsistent with the requirements of "constructing a multi-level education system and strengthening the scientificity of evaluation" proposed in the New Generation Artificial Intelligence Development Plan, restricting the quality upgrading of AI education in higher vocational colleges.

Against this backdrop, based on the data-driven concept, this study aims to construct a full-process evaluation system covering pre-class, in-class and post-class stages. By integrating the theory of multiple intelligences with intelligent algorithm technology, this research breaks the single and static limitations of traditional course evaluation. The research outcomes can provide a practical paradigm for higher vocational colleges to implement national AI education policies and improve course quality, and facilitate the cultivation of compound talents with both AI literacy and professional vocational capabilities.

1.1. Subsection title

There may be three levels of headings, used by selecting style Heading 1, Heading 2 or Heading 3. For capitalization, use initial capitals in chapter titles and headings, with the exception of prepositions and articles (e.g., "the", "and").

1.1.1. Subsubsection title

When enumeration is needed, Authors are directed to use bulleted (full round bullets "●") lists. Only if the items ranking or ordering has some meaning (e.g., as in enumerating the steps of an algorithm), the numbered lists could be used.

An example of unnumbered list:

2. Research Foundation and Methods

2.1. Research Foundation

This study refers to the Network Course Evaluation Specification (CELTS22) issued by the Technical Standard Committee for Educational Informatization of the Ministry of Education. The specification covers multi-dimensional evaluation structures including platform quality, teaching content and teaching design, laying a solid foundation for the construction of the “online + offline” hybrid evaluation indicators in this research. In combination with the particularity of the research object, this study integrates the ideological and political integration demand of the “red finance and economics” curriculum in Sichuan Vocational College of Finance and Economics, as well as the three-in-one construction goal of “knowledge chain + skill chain + ideological and political chain” for higher vocational AI courses. On this basis, it further clarifies the unique requirements for professional adaptability and ideological and political connotation of the evaluation system, providing a targeted basis for indicator design.

2.2. Research Methods

Adopting a data-driven technical framework, this study applies two core research methods to ensure the scientificity and accuracy of the evaluation system.

First, the principal component analysis (PCA) method is utilized. Considering the phased limitations of real survey data acquisition, the sample size of simulated questionnaire data is determined based on the statistical sample size estimation formula and adjusted according to the proportion of three types of respondents including experts, teachers and students. Taking

the smart classroom evaluation system for higher vocational colleges proposed by Zhou Jianjun as the basic framework, preliminary evaluation indicators are optimized through expert surveys. The PCA method is then used for indicator dimensionality reduction, which retains the main information of original data while simplifying the evaluation system and improving its practical application effectiveness.

Second, intelligent optimization algorithms are introduced. The grey wolf optimization (GWO) algorithm is adopted to optimize the core hyperparameters of the random forest model (including the number of decision trees, maximum tree depth, and minimum sample segmentation number). The weight of each evaluation indicator is determined by minimizing the root mean square error (RMSE), so as to enhance the accuracy and operational efficiency of the evaluation model and guarantee the scientificity of evaluation results. According to the common parameter range of GWO applied in existing higher vocational course evaluation studies, pre-experimental results show that the model converges after 20 iterations, and additional iterations cannot significantly reduce the error. Therefore, 20 iterations and 10 wolf pack individuals (population size) are set in this study, which balances computational efficiency and optimization accuracy and ensures sufficient optimization searching.

3. Construction of the Evaluation System for General Artificial Intelligence Courses in Higher Vocational Education

To construct an AI-era-adaptive, full-process evaluation system for general AI courses in higher vocational colleges, this study takes the comprehensive evaluation method as the theoretical framework and integrates the theory of multiple intelligences. Relying on artificial intelligence technology to realize data collection, analysis and dynamic feedback, the system achieves comprehensive, dynamic and personalized evaluation of students' learning outcomes.

3.1. Preliminary Evaluation Indicator System

Based on literature review and referring to the smart classroom evaluation indicators obtained via the Delphi method and reliability and validity test in Zhou Jianjun's 2024 research on smart classroom evaluation systems for higher vocational colleges in the AI era, this study revises indicators through expert demonstration and constructs a preliminary evaluation indicator system covering pre-class, in-class and post-class dimensions. The system includes 8 first-level indicators (teaching preparation, teaching environment, information literacy,

teaching ability, classroom adaptability, classroom interaction, classroom regulation and teaching effect) and corresponding secondary indicators, with detailed items shown in Table 1.

Table 1. Preliminary Evaluation Indicator System for General Artificial Intelligence Courses in Higher Vocational Education.

No.	First-level Indicator	Indicator Item	No.	First-level Indicator	Indicator Item
A1	Teaching Preparation	Integrity of teaching structure design	A19	Classroom Interaction	Number of online teacher-student interactions
A2		Adequacy of teaching materials and resources	A20		Effectiveness of learning suggestions
A3		Completeness of teaching syllabus design	A21		Rationality of Q&A session arrangement
A4		Completeness of teaching facilities	A22		Perfection of student feedback channels
A5	Teaching Environment	Completeness of multimedia configuration	A23	Classroom Regulation	Targetedness of teaching content
A6		Harmony of teacher-student relationship	A24		Flexibility of teaching methods
A7		Stability of network platform operation	A25		Visualization of teaching process
A8	Information Literacy	Mastery of new digital teaching technologies	A26	Teaching Effect	Completion rate of online quizzes
A9		Proficiency in using digital teaching aids	A27		Improvement of students' independent learning ability
A10		Ability to collect digital resources	A28		Accuracy rate of student task completion
A11	Teaching Ability	Ability to process teaching materials	A29		Scores of students' post-class assessments
A12		Standardization of teaching organization	A30		Improvement of students' comprehensive literacy
A13		Effectiveness of teaching organization	A31		Development of students' problem-solving ability
A14		Accuracy of curriculum ideological and political understanding	A32		Development of students' innovative thinking
A15	Classroom Adaptability	Adaptability to new technologies	A33		Development of students' teamwork ability
A16		Acceptance of new teaching methods	A34		Improvement of students' language expression ability
A17		Adaptability to new teaching environments	A35		Improvement of students' introspective reflection ability

A18		Adaptability to students' learning states	A36		Growth and progress range of students
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3.2. Questionnaire Design and Data Collection

To verify and optimize the preliminary evaluation indicator system, this study designs a formal questionnaire distributed to professional experts, frontline teachers and students engaged in higher vocational general AI course teaching. A five-point Likert scale (1 to 5 points, corresponding to “extremely unimportant” to “extremely important” respectively) is adopted to score the importance of each indicator.

The questionnaire design follows the principles of comprehensiveness, scientificity and operability. In terms of comprehensiveness, the questionnaire fully covers all evaluation dimensions of higher vocational general AI course teaching. In terms of scientificity, the questionnaire content is revised for multiple rounds by combining the Delphi method and expert interviews to ensure the professionalism and validity of indicators. In terms of operability, all questions are designed clearly and specifically to facilitate respondents' filling and subsequent quantitative data analysis.

Data collection adopts a combined method of online questionnaire distribution and offline interviews to ensure the breadth and depth of research data, and the collected data serves as the basic data for subsequent PCA and intelligent algorithm analysis.

3.3. PCA Dimensionality Reduction

To extract core key indicators from the preliminary system and improve evaluation efficiency and accuracy, this study conducts dimensionality reduction on simulated questionnaire data via PCA. PCA can convert high-dimensional original data into a set of orthogonal low-dimensional variables (principal components) while retaining most valid information of the original data.

This study implements PCA analysis using the scikit-learn library in Python. First, simulated data is standardized to eliminate the dimensional differences between various indicators. Then, the explained variance ratio and cumulative explained variance ratio of each principal component are calculated to determine the number of retained principal components.

3.3.1. Explained Variance Ratio and Cumulative Explained Variance

The results of PCA analysis on the explained variance ratio and cumulative explained variance are as follows:

Explained Variance Ratio:

```
[0.04864629 0.04372884 0.0418546 0.04000878 0.03918137 0.03798454  
0.03687013 0.03599147 0.0356437 0.03415733 0.03396485 0.03293688  
0.0318472 0.03101717 0.03001313 0.02943617 0.02898938 0.02871376  
0.02827191 0.02656802 0.02605609 0.02518233 0.02486386 0.02422207  
0.023231 0.02291812 0.02242318 0.02120343 0.0209201 0.0202677  
0.01945949 0.01899344 0.01851611 0.01591755]
```

Cumulative Explained Variance:

```
[0.04864629 0.09237513 0.13422973 0.17423851 0.21341988 0.25140443  
0.28827455 0.32426602 0.35990972 0.39406705 0.4280319 0.46096879  
0.49281599 0.52383316 0.55384629 0.58328245 0.61227183 0.64098559  
0.6692575 0.69582553 0.72188162 0.74706395 0.7719278 0.79614988  
0.81938088 0.842299 0.86472218 0.8859256 0.90684571 0.92711341  
0.94657289 0.96556633 0.98408245 1. ]
```

According to the cumulative explained variance results, the first 29 principal components achieve a cumulative explained variance of 90.68%, exceeding the 90% information retention threshold. Therefore, this study retains the first 29 principal components, reducing the original 36 evaluation indicators to 29 core components, which realizes data dimensionality reduction while retaining most valid original information.

3.3.2. Key Indicator Extraction

By analyzing the loading value of original indicators in each principal component, the indicators with the largest contribution to each principal component are identified as key evaluation indicators. This study selects the top 3 indicators with the maximum absolute loading values for each principal component, and the extracted key indicators are as follows:

Key Indicators for each Principal Component:

```
PC1: ['A18', 'A21', 'A35']  
PC2: ['A15', 'A32', 'A34']  
PC3: ['A23', 'A14', 'A25']  
PC4: ['A10', 'A36', 'A7']  
PC5: ['A33', 'A4', 'A3']  
PC6: ['A34', 'A14', 'A35']  
PC7: ['A24', 'A31', 'A13']  
PC8: ['A12', 'A28', 'A32']  
PC9: ['A7', 'A30', 'A6']
```

```
PC10: ['A31', 'A22', 'A2']
PC11: ['A17', 'A30', 'A4']
PC12: ['A19', 'A30', 'A29']
PC13: ['A7', 'A16', 'A6']
PC14: ['A28', 'A5', 'A16']
PC15: ['A5', 'A31', 'A7']
PC16: ['A24', 'A6', 'A17']
PC17: ['A33', 'A29', 'A1']
PC18: ['A4', 'A25', 'A29']
PC19: ['A20', 'A8', 'A6']
PC20: ['A21', 'A18', 'A30']
PC21: ['A23', 'A24', 'A13']
PC22: ['A20', 'A24', 'A32']
PC23: ['A16', 'A5', 'A11']
PC24: ['A34', 'A10', 'A7']
PC25: ['A17', 'A32', 'A25']
PC26: ['A8', 'A20', 'A13']
PC27: ['A32', 'A36', 'A13']
PC28: ['A33', 'A29', 'A21']
PC29: ['A14', 'A5', 'A15']
```

The above key indicators retain the core effective information of the original evaluation system, which serves as the core basis for subsequent evaluation model construction and significantly improves the efficiency and accuracy of course evaluation.

3.4. Random Forest Model Optimization Based on Intelligent Optimization Algorithm

To further enhance the accuracy and generalization ability of the teaching evaluation model, this study adopts the GWO algorithm to optimize the hyperparameters of the random forest model, and calculates the weight of each evaluation indicator based on the optimized model. As an ensemble learning algorithm, random forest possesses high prediction accuracy and robustness, but its performance is highly dependent on hyperparameter settings. The GWO algorithm, as a novel swarm intelligence optimization algorithm, has the advantages of fast convergence and strong optimization ability, which is suitable for solving complex parameter optimization problems.

This study takes four core hyperparameters of the random forest model (`n_estimators`, `max_depth`, `min_samples_split`, `min_samples_leaf`) as the optimization objectives of GWO, and adopts RMSE as the fitness function to search for the optimal hyperparameter combination that minimizes model prediction error. In the simulation experiment, 20 iterations and 10 wolf pack individuals are set for optimization.

3.4.1. Optimization Results

The optimal hyperparameter combination of the random forest model and the optimal test-set RMSE obtained by GWO optimization are as follows:

```
Optimized Random Forest Hyperparameters (n_estimators, max_depth,  
min_samples_split, min_samples_leaf):  
[150, 18, 4, 2]
```

```
Best RMSE on test set:  
0.7351471057220035
```

The optimized random forest model achieves a low RMSE on the test set, verifying that the GWO algorithm effectively improves the prediction performance of the evaluation model and enables more accurate prediction of teaching evaluation results.

3.4.2. Weight of Evaluation Indicators

The importance (weight) of each evaluation indicator is extracted based on the optimized random forest model, which reflects the contribution degree of each indicator to the overall teaching evaluation result. The indicator weight ranking is shown as follows:

```
Feature Importances (Weights) from Optimized Random Forest:  
A31:0.100235, A32:0.093814, A30:0.088185, A36:0.077068, A9:0.058668,  
A10:0.051579, A14:0.048769, A29:0.046989, A11:0.033493, A8:0.030842,  
A28:0.025826, A13:0.023536, A23:0.022692, A1:0.022456, A2:0.022168,  
A20:0.021419, A25:0.019149, A12:0.017272, A19:0.015944, A3:0.015445,  
A4:0.013204, A5:0.012880, A18:0.012809, A21:0.012591, A33:0.012540,  
A24:0.012318, A22:0.012186, A34:0.012001, A6:0.011919, A17:0.011830,  
A7:0.011549, A15:0.010452, A16:0.009098, A35:0.009076
```

The weight results indicate that student problem-solving ability (A31), innovative thinking development (A32), comprehensive literacy improvement (A30) and growth progress range (A36) have the highest weights, playing a dominant role in the overall teaching evaluation.

The results verify the effectiveness of the GWO-optimized model and provide a quantitative basis for the construction of the final evaluation system.

3.5. Final Evaluation System

Combining the key indicators screened by PCA dimensionality reduction and the indicator weights optimized by the GWO-random forest model, this study constructs the final evaluation system for higher vocational general AI courses. The system simplifies the evaluation process while retaining core evaluation information, ensuring the scientificity, accuracy and practicability of evaluation. The detailed structure of the final evaluation system is shown in Table 2.

Table 2. Final Evaluation System for General Artificial Intelligence Courses in Higher Vocational Education.

First-level Indicator	Secondary Key Indicator	Weight	Evaluation Description
Teaching Effect	Student problem-solving ability (A31)	0.100235	Evaluate students' ability to apply artificial intelligence knowledge to analyze, model and solve practical problems, aiming to cultivate students' complex problem-solving competence and innovative spirit.
	Student innovative thinking development (A32)	0.093814	Evaluate the unique insights and innovative schemes demonstrated by students in AI application practice, reflecting the cultivation of creative intelligence and encouraging students to explore and break through conventional thinking.
	Student comprehensive literacy improvement (A30)	0.088185	Evaluate students' comprehensive development in AI ethics, social responsibility, critical thinking and innovative awareness, reflecting the all-round cultivation of multiple intelligences and the shaping of patriotism and social responsibility.
	Student growth and progress range (A36)	0.077068	Evaluate the learning progress and ability improvement of students during the learning process, adhering to the value-added evaluation concept that focuses on individual growth.
	Student post-class assessment scores (A29)	0.046989	Reflect students' knowledge consolidation and ability migration level after class, as well as their understanding and recognition of artificial intelligence ethics norms.
	Student task completion accuracy rate (A28)	0.025826	Reflect students' mastery of professional knowledge and practical operation skills, and cultivate students' rigorous and pragmatic scientific spirit in task completion.

	Student teamwork ability (A33)	0.012540	Evaluate students' ability to communicate and cooperate effectively in group projects, reflecting the cultivation of interpersonal intelligence and collective spirit.
	Student language expression ability (A34)	0.012001	Evaluate students' ability to accurately and clearly express AI-related concepts and viewpoints, integrating scientific literacy and humanistic literacy cultivation.
	Student introspective reflection ability (A35)	0.009076	Evaluate students' ability to summarize, reflect and optimize learning methods based on their own learning process and outcomes, cultivating self-awareness and self-improvement awareness.
Information Literacy	Proficiency in using digital teaching aids (A9)	0.058668	Evaluate teachers' ability to flexibly apply AI software and hardware teaching aids to enrich teaching forms and stimulate students' interest in emerging technologies.
	Ability to collect digital resources (A10)	0.051579	Evaluate teachers' ability to acquire, screen and integrate cutting-edge AI digital resources for teaching, and cultivate students' critical thinking and information discrimination ability.
	Mastery of new digital teaching technologies (A8)	0.030842	Evaluate teachers' proficiency in applying various digital teaching tools, improving teaching efficiency and classroom attraction, and setting a model for students' digital learning.
Teaching Ability	Accuracy of curriculum ideological and political understanding (A14)	0.048769	Evaluate teachers' ability to accurately grasp the connotation of curriculum ideological and political education and integrate it into AI teaching imperceptibly to realize simultaneous knowledge transmission and value guidance.
	Teaching material processing ability (A11)	0.033493	Evaluate teachers' ability to restructure teaching content based on AI industry development and ideological and political teaching requirements, ensuring the integration of professional knowledge and moral education.
	Effectiveness of teaching organization (A13)	0.023536	Evaluate the rationality of teaching activity design, which can effectively stimulate students' learning initiative, promote knowledge internalization and shape correct values.
	Standardization of teaching organization (A12)	0.017272	Evaluate teachers' classroom rhythm control ability and standardized teaching organization level, to build a positive and efficient classroom atmosphere.
Classroom Regulation	Targetedness of teaching content (A23)	0.022692	Evaluate teachers' ability to adjust teaching content according to students' learning differences and progress, realizing

			differentiated and personalized teaching for all students.
	Visualization of teaching process (A25)	0.019149	Evaluate the application of AI technology in teaching process visualization, which facilitates teachers and students to track learning trajectories and improve teaching transparency and efficiency.
	Flexibility of teaching methods (A24)	0.012318	Evaluate teachers' ability to flexibly adopt diversified teaching methods to adapt to different students' learning styles and improve teaching pertinence and effectiveness.
Teaching Preparation	Integrity of teaching structure design (A1)	0.022456	Evaluate the rationality of curriculum objective setting and teaching logic design, reflecting the cutting-edge nature of AI courses and the integration degree of ideological and political education.
	Adequacy of teaching materials and resources (A2)	0.022168	Evaluate the richness and diversity of online and offline teaching resources, and examine the value of resources in supporting professional teaching and ideological and political education.
	Completeness of teaching syllabus design (A3)	0.015445	Evaluate the standardization of syllabus design and the rationality of teaching link arrangement, clarifying the path of professional ability and comprehensive literacy cultivation.
	Completeness of teaching facilities (A4)	0.013204	Evaluate the matching degree of teaching software and hardware facilities with course teaching requirements, providing reliable support for AI practical teaching.
Classroom Interaction	Effectiveness of learning suggestions (A20)	0.021419	Evaluate teachers' ability to put forward targeted personalized learning suggestions based on students' learning data, practicing the student-centered teaching concept.
	Number of online teacher-student interactions (A19)	0.015944	Reflect the activity of online classroom interaction, promote two-way communication between teachers and students, and realize mutual promotion of teaching and learning.
	Rationality of Q&A session arrangement (A21)	0.012591	Evaluate the rational design of classroom Q&A links, which guides students to think independently and question actively, and cultivates critical thinking ability.
	Perfection of student feedback channels (A22)	0.012186	Evaluate the completeness of the multi-dimensional student feedback mechanism, which timely captures students' learning difficulties and demands to support teaching optimization.

Classroom Adaptability	Adaptability to students' learning states (A18)	0.012809	Evaluate teachers' ability to perceive students' learning status in real time and solve learning problems timely, reflecting teachers' educational responsibility and teaching wisdom.
	Adaptability to new teaching environments (A17)	0.011830	Evaluate the adaptability of teachers and students to smart classrooms, virtual simulation laboratories and other new teaching environments, cultivating lifelong learning and adaptive ability.
	Adaptability to new technologies (A15)	0.010452	Evaluate the ability of teachers and students to quickly adapt to emerging AI technologies and tools, adapting to the iterative development of the intelligent industry.
	Acceptance of new teaching methods (A16)	0.009098	Evaluate the acceptance of teachers and students to hybrid teaching, project-based learning and other innovative teaching modes, reflecting the openness to educational innovation.
Teaching Environment	Completeness of multimedia configuration (A5)	0.012880	Evaluate the configuration level and operational stability of classroom multimedia equipment, supporting the implementation of diversified intelligent teaching modes.
	Harmony of teacher-student relationship (A6)	0.011919	Evaluate the harmony of classroom teacher-student interaction, build a positive educational atmosphere, and realize the imperceptible infiltration of ideological and political education.
	Stability of network platform operation (A7)	0.011549	Evaluate the operational stability of online teaching platforms, ensuring the smooth progress of online learning and resource sharing, and supporting the dissemination of ideological and political teaching resources.

The constructed evaluation system not only focuses on students' knowledge mastery and professional skill improvement, but also emphasizes the cultivation of autonomous learning ability, practical innovation ability and comprehensive literacy in the AI era. The quantitative weight setting of each indicator realizes objective and accurate teaching effect evaluation, providing reliable data support for teaching optimization and students' personalized development in higher vocational colleges.

4. Conclusion and Prospect

This study focuses on the quality evaluation of general AI courses in higher vocational colleges and constructs a full-process evaluation framework covering pre-class, in-class and post-class stages. Integrating the theory of multiple intelligences and data-driven intelligent optimization technology, the framework effectively solves the defects of traditional evaluation

systems such as single evaluation mode, delayed feedback, insufficient dynamic monitoring and lack of personalized evaluation. The research outcomes can provide theoretical support and practical guidance for the teaching reform and quality improvement of general AI courses in higher vocational colleges, facilitate the personalized development and comprehensive ability cultivation of students, and promote the adaptation of higher vocational talent training quality to the demand for AI-era compound technical talents.

This study still has certain limitations. First, the indicator weight calculation does not consider the differences between students of different majors (such as finance and economics majors and engineering majors), and the indicator weight system can be optimized by major categories in subsequent research. Second, the current evaluation system lacks the evaluation perspective of enterprises, and enterprise tutors can be introduced to participate in evaluation in follow-up studies to improve the practical applicability of the system.

Future research can be carried out from four aspects: conducting large-scale empirical research with real teaching data to verify the effectiveness and applicability of the evaluation system; developing intelligent automatic evaluation tools and platforms based on the constructed evaluation framework; exploring the specific application path of evaluation results in teaching optimization and students' learning strategy adjustment; and discussing the ethical risks of AI technology in educational evaluation, so as to put forward solutions for ensuring the fairness, transparency and credibility of intelligent evaluation.

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